



GREENMIND AI

Smart Plant Disease Detection & Botanical Decision Support System

RESEARCH PROJECT PROPOSAL

Research Field:	Agronomic AI, Computer Vision & Expert Systems
Project Duration:	36 Months (3-Year Phased Timeline)
Proposed For:	National Agricultural R&D Grant Initiative
Date of Proposal:	June 11, 2026



i) Project Title

GreenMind AI: A Hybrid Deep Learning and Generative AI Ecosystem for Real-time Plant Disease Detection and Botanical Decision Support.

ii) Introduction

1. Origin of the Research Problem

Global agricultural production and local home gardening face severe threats from plant pathogens. According to the FAO, plant diseases and pests destroy 20% to 40% of global crop yields annually, costing the world economy over \$220 billion. Early identification is key to mitigating spread, yet access to expert botanists or plant pathologists is highly restricted, particularly in rural and developing regions. Existing mobile plant care applications rely on static databases and stand-alone classification models. While they can label a disease, they fail to provide customized, context-aware care guidelines or engage in interactive Q&A. This leads to user frustration, misapplication of chemical treatments, and high abandonment rates. GreenMind AI originates from the need to bridge this gap by offering a hybrid framework that pairs instant classification with the deep reasoning of Generative AI.

2. Interdisciplinary Relevance

This research is highly interdisciplinary, merging four critical pillars:

- Phytopathology: Analyzing crop-specific pathogens, symptoms, and care models.
- Computer Vision: Designing and training highly optimized CNNs for leaf disease classification.
- Natural Language Processing: Employing Large Language Models to generate tailored botanical care plans.
- Systems Engineering: Implementing secure mobile applications and serverless FastAPI backends.

3. Review of Research and Development in the Subject

A. International Status

Globally, researchers have leveraged deep learning architectures (like ResNet, MobileNet, and EfficientNet) trained on the public PlantVillage dataset (54,000+ images) to classify crop leaf diseases. International commercial apps like Plantix have successfully deployed CNN models for remote detection. However, these systems operate as closed-source diagnostic silos. They do not utilize Generative AI for customized conversation, nor do they support context-aware parameters such as weather forecasts or custom soil properties during treatment planning.

B. National Status

In India, digital platforms such as mKisan and e-NAM provide general agricultural advisories, but lack instant, image-based plant diagnostics. While regional universities have trained CNNs on local crop varieties (such as paddy, wheat, and cotton), their deployments are often heavy web services with high latencies. There is a clear absence of mobile-first, low-bandwidth solutions that translate predictions into natural, localized languages with automated PDF reporting.

C. Significance of the Study

GreenMind AI overcomes these limitations by combining a lightweight convolutional neural network with generative large language models. The custom model classifies leaf diseases with high speed, and the backend LLM translates the diagnostic label into a complete treatment plan. The study introduces a fully automated reporting system that formats plant diagnostics into styled, professional PDFs, making expert agricultural knowledge accessible to anyone with a smartphone.



iii) Objectives of the Project

- **O1: Deep Learning Classification:** Train and optimize a lightweight CNN model (MobileNetV2 backbone) targeting 38 crop-disease pairs to achieve >92% classification accuracy.
- **O2: Hybrid AI Inference:** Implement a service-oriented backend that orchestrates image classification via the CNN and feeds the outputs to GPT-4o to construct a tailored care script.
- **O3: Cross-platform Mobile Client:** Develop a modern, animated Flutter app supporting offline caching, real-time diagnostic history sync, and interactive chat support.
- **O4: Automated PDF Dispatch:** Build a serverless FPDF2 reporting engine and SMTP mail pipeline to automatically compile and email plant health records to users.
- **O5: Scalable Cloud Architecture:** Establish a serverless FastAPI backend on Vercel connected to Supabase for secure authentication, history database storage, and file hosting.

iv) Methodology

The project follows a modular, service-oriented system design. The core methodology is split into four phases:

1. Data Curation & CNN Model Training:

Gather and clean agricultural leaf images (PlantVillage dataset + field collections). Conduct augmentations (rotation, shear, brightness shift) to avoid overfitting. Train a TensorFlow MobileNetV2 classifier. Export the model to a compact format for fast inference.

2. Hybrid Inference Pipeline (Dual-Layer Intelligence):

When a user uploads a leaf image, the FastAPI backend routes it to the CNN classifier. If the confidence is above 70%, the backend triggers the GPT-4o API, providing it with the crop name, identified disease, confidence, and metadata. GPT-4o then compiles the possible cause and detailed treatment plan.

3. Cross-Platform App & Real-Time Sync:

Build a Flutter mobile client incorporating provider-based state management. Store history locally using sqlite or hive. Synchronize records automatically with Supabase (PostgreSQL) when an active internet connection is detected. Provide text-to-speech (TTS) features for accessibility.

4. Automated PDF & SMTP Dispatch:

A background task compiles the diagnosis and treatment plan. A class inheriting from FPDF formats the text, appends branding, and generates a PDF document. The SMTP client then dispatches this PDF directly to the user's email.

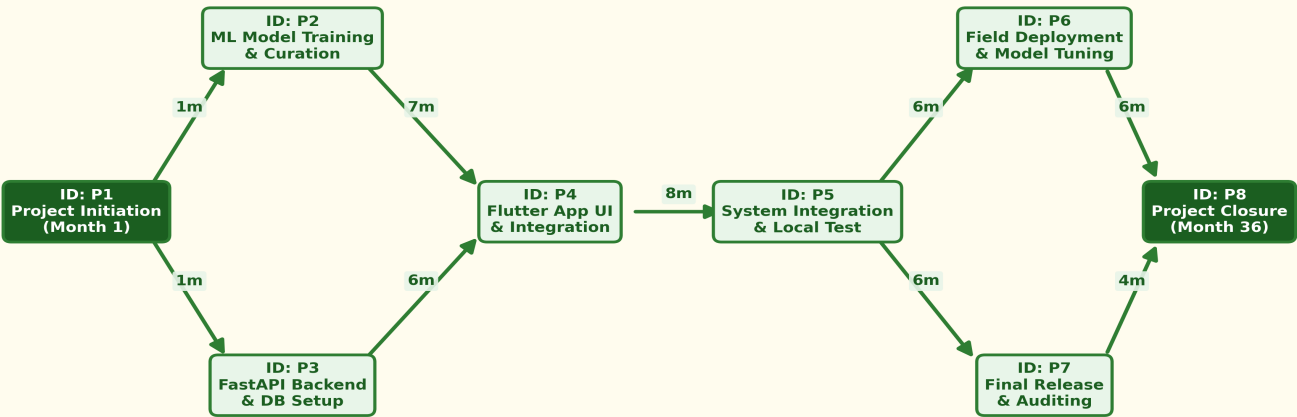


v) Year wise Plan of Work & Targets

Task ID	Milestone / Target Deliverable	Timeline
P1.1	Dataset curation and plant leaf image preprocessing	Months 1-3
P1.2	CNN model architecture design, training, and testing	Months 4-7
P1.3	FastAPI backend architecture and Supabase database setup	Months 8-12
P2.1	Flutter app core UI development and camera module integration	Months 13-16
P2.2	Hybrid dual-inference engine (CNN + GPT API) integration	Months 17-20
P2.3	FPDF2-based automated PDF reporting engine and SMTP setup	Months 21-24
P3.1	End-to-end client-server integration and local testing	Months 25-28
P3.2	Field trials with local farming cooperatives and users	Months 29-32
P3.3	Retraining model on field data, security audit, and release	Months 33-36

PERT Network Diagram (36-Month Project Path)

GreenMind AI: PERT Network Diagram (36-Month Timeline)





vi) Financial Estimates / Budget

Sr. No.	Items	Est. Expenditure (INR)
i	Hiring Services	0
ii	Field Work & Travel	7,000
iii	Chemical & Glassware	0
iv	Contingency (Including special needs)	10,000
v	Books and Journals	15,000
vi	Software Licenses / Cloud Services	8,000
vii	Dataset Collection & Data Annotation	5,000
viii	Computer Peripherals & Storage Devices	6,000
ix	Internet & Communication Charges	3,000
x	Project Demonstration & Prototype Development	6,000
xi	Miscellaneous (Printing, Documentation)	5,000
Total	Total in INR	65,000